

Experiment 20 :

Implement a Basic Information Retrieval System Using TF-IDF for Document Ranking

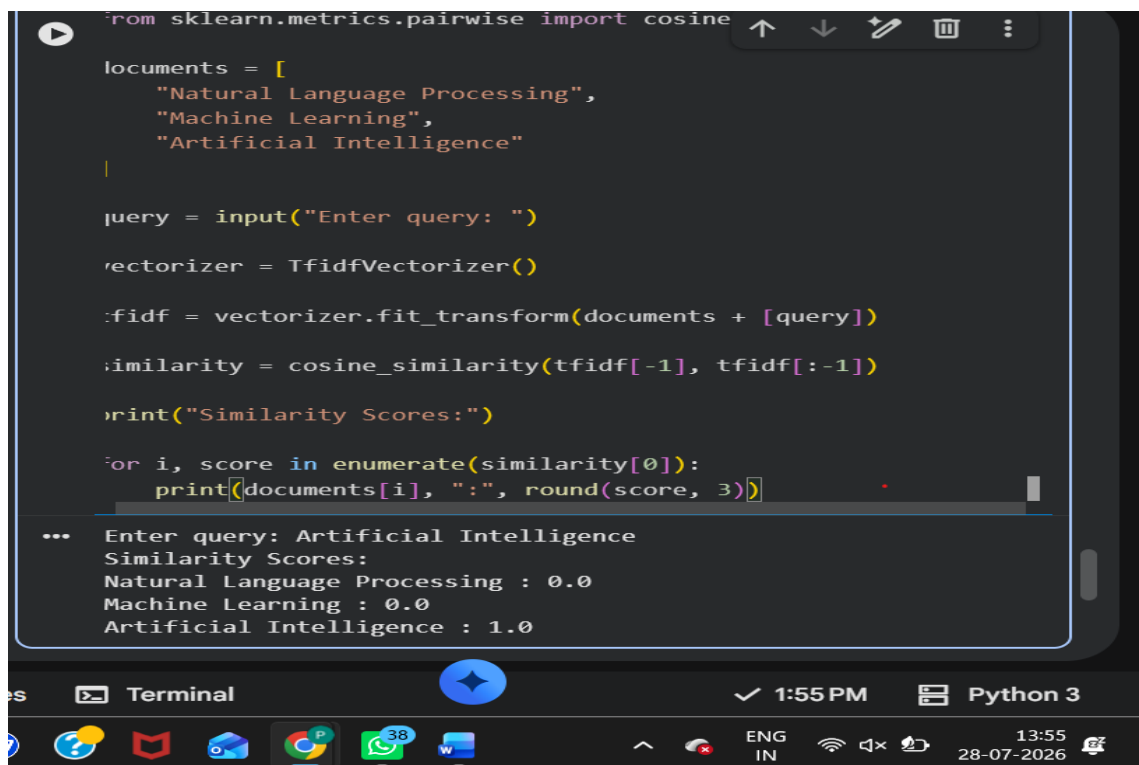
Code:

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

documents = [
    "Natural Language Processing",
    "Machine Learning",
    "Artificial Intelligence"
]

query = input("Enter query: ")
vectorizer = TfidfVectorizer()
tfidf = vectorizer.fit_transform(documents + [query])
similarity = cosine_similarity(tfidf[-1], tfidf[:-1])
print("Similarity Scores:")
for i, score in enumerate(similarity[0]):
    print(documents[i], ":", round(score, 3))
```

Output



```
from sklearn.metrics.pairwise import cosine
documents = [
    "Natural Language Processing",
    "Machine Learning",
    "Artificial Intelligence"
]

query = input("Enter query: ")
vectorizer = TfidfVectorizer()
tfidf = vectorizer.fit_transform(documents + [query])
similarity = cosine_similarity(tfidf[-1], tfidf[:-1])
print("Similarity Scores:")
for i, score in enumerate(similarity[0]):
    print(documents[i], ":", round(score, 3))

... Enter query: Artificial Intelligence
Similarity Scores:
Natural Language Processing : 0.0
Machine Learning : 0.0
Artificial Intelligence : 1.0
```